

Ex.no:5B	DATA VISUALIZATION USING MATPLOTLIB

Aim:

To visualize the Life Expectancy dataset features and distributions using line charts, bar charts, pie charts, histograms, scatter plots, and box plots using Matplotlib.

Algorithm:

ALGORITHM 1: LINE CHART

Objective: To visualize the average Life Expectancy across different years.

Steps

1. Load the Life Expectancy dataset.
2. Group the data by Year and calculate the average Life expectancy.
3. Arrange the yearly averages in chronological order.
4. Plot the average Life expectancy using a line chart.
5. Add title, axis labels, and grid.
6. Display the graph.

Interpretation

The line chart shows how average Life Expectancy varies across different years and helps identify overall trends.

```

year_avg = df.groupby("Year")["Life expectancy"].mean()
plt.figure(figsize=(6,4))
plt.plot(year_avg.index, year_avg.values, marker='o')
plt.title("Year vs Average Life Expectancy")
plt.xlabel("Year")
plt.ylabel("Average Life Expectancy")
plt.grid(True)
plt.show()

```

ALGORITHM 2: BAR CHART

Objective: To compare the average Life Expectancy between different development statuses.

Steps

1. Select the Status column.
2. Group the data by Status and calculate the average Life expectancy.
3. Create a bar chart using the status categories and their average Life Expectancy.
4. Add title and axis labels.
5. Display the chart.

Interpretation

The bar chart provides a comparison of average Life Expectancy between the development status categories in the dataset.

```
status_avg = df.groupby("Status")["Life expectancy"].mean()
plt.figure(figsize=(6,4))
plt.bar(status_avg.index, status_avg.values)
plt.title("Average Life Expectancy by Status")
plt.xlabel("Status")
plt.ylabel("Average Life Expectancy")
plt.show()
```

ALGORITHM 3: PIE CHART

Objective: To visualize the distribution of development status in the Life Expectancy dataset.

Steps

1. Select the Status column.
2. Count the occurrences of each development status.
3. Calculate their proportions.
4. Create a pie chart.
5. Display percentage labels.
6. Add a suitable title.

Interpretation

The pie chart shows the percentage contribution of each customer type to the total bookings.

```
status_counts = df["Status"].value_counts()
plt.figure(figsize=(6,6))
plt.pie(status_counts, labels=status_counts.index, autopct="%1.1f%%",
startangle=90)
```

```
plt.title("Development Status Distribution")  
plt.show()
```

ALGORITHM 4: HISTOGRAM

Objective: To visualize the distribution of Life Expectancy.

Steps

1. Select the Life expectancy column.
2. Remove missing Life expectancy values.
3. Divide the Life Expectancy values into intervals or bins.
4. Create a histogram using Matplotlib.
5. Add title and axis labels.
6. Display the histogram.

Interpretation

The histogram shows the frequency distribution of Life Expectancy values and helps understand the spread of the target variable.

```
plt.figure(figsize=(6,4))  
plt.hist(df["Life expectancy"].dropna(), bins=10)  
plt.title("Life Expectancy Distribution")  
plt.xlabel("Life Expectancy")  
plt.ylabel("Frequency")  
plt.show()
```

ALGORITHM 5: SCATTER PLOT

Objective: To analyze the relationship between Schooling and Life Expectancy.

Steps

1. Select Schooling as the X-axis variable.
2. Select Life expectancy as the Y-axis variable.
3. Plot the observations using a scatter plot.
4. Add title and axis labels.
5. Display the graph.

Interpretation

The scatter plot helps identify the visible relationship between Schooling and Life Expectancy in the dataset.

```
plt.figure(figsize=(6,4))
plt.scatter(df["Schooling"], df["Life expectancy"])
plt.title("Schooling vs Life Expectancy")
plt.xlabel("Schooling")
plt.ylabel("Life Expectancy")
plt.show()
```

ALGORITHM 6: BOX PLOT

Objective: To visualize the distribution and identify possible outliers in Life Expectancy.

Steps

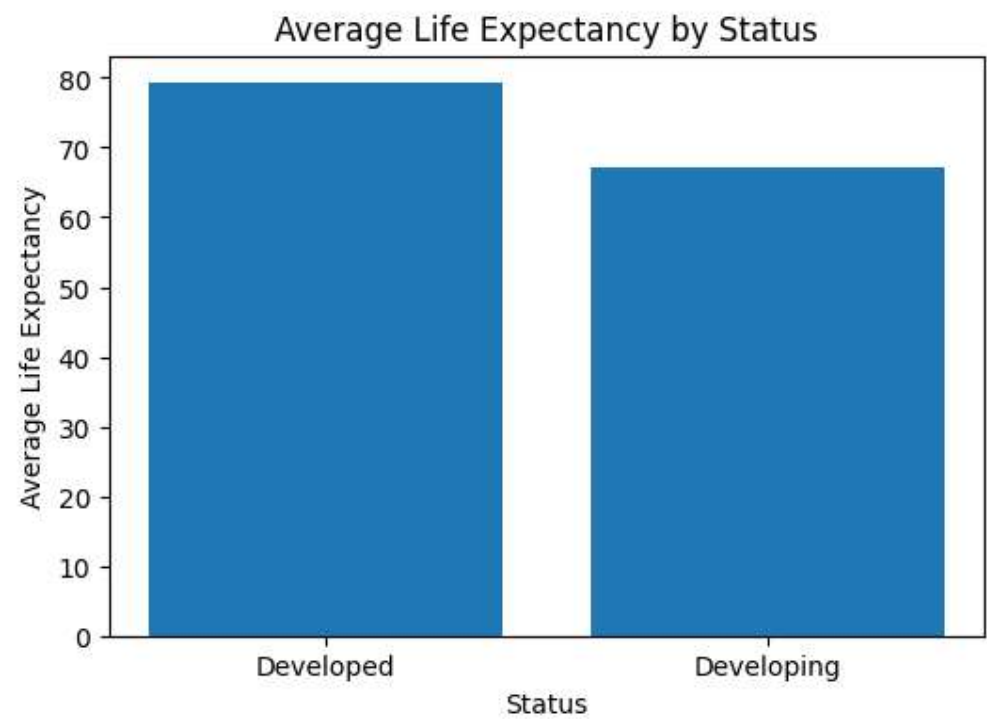
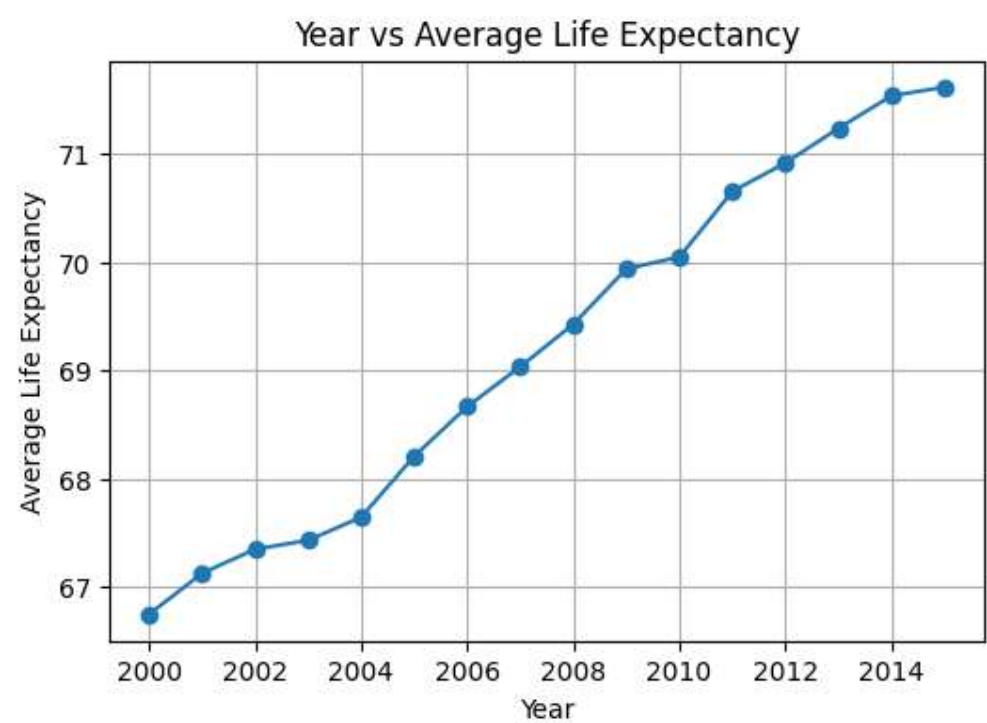
1. Select the Life expectancy column.
2. Remove missing values if present.
3. Create a box plot using Matplotlib.
4. Add a suitable title and axis label.
5. Display the plot.

Interpretation

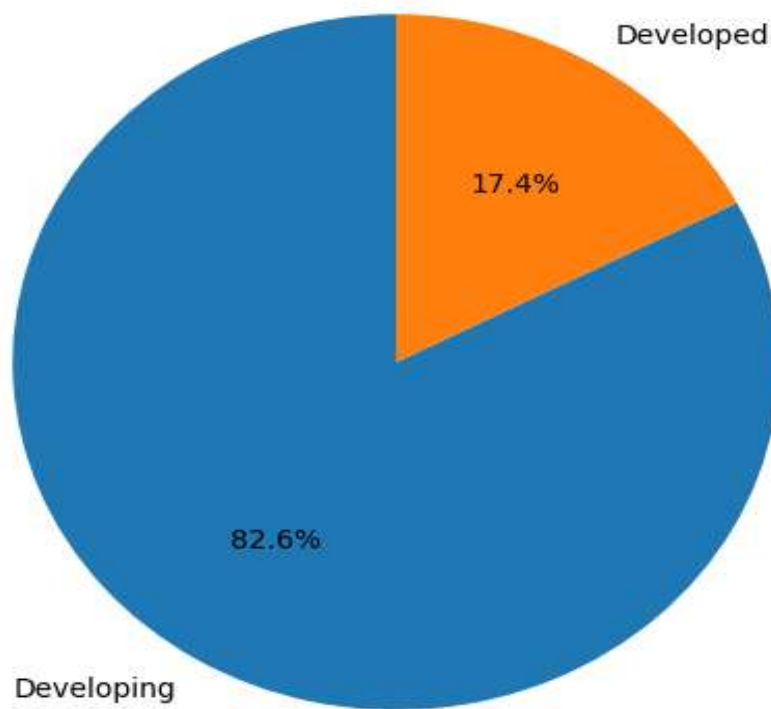
The box plot shows the central distribution of Life Expectancy and helps identify unusually high or low values.

```
plt.figure(figsize=(4,5))
plt.boxplot(df["Life expectancy"].dropna())
plt.title("Life Expectancy Box Plot")
plt.ylabel("Life Expectancy")
plt.show()
```

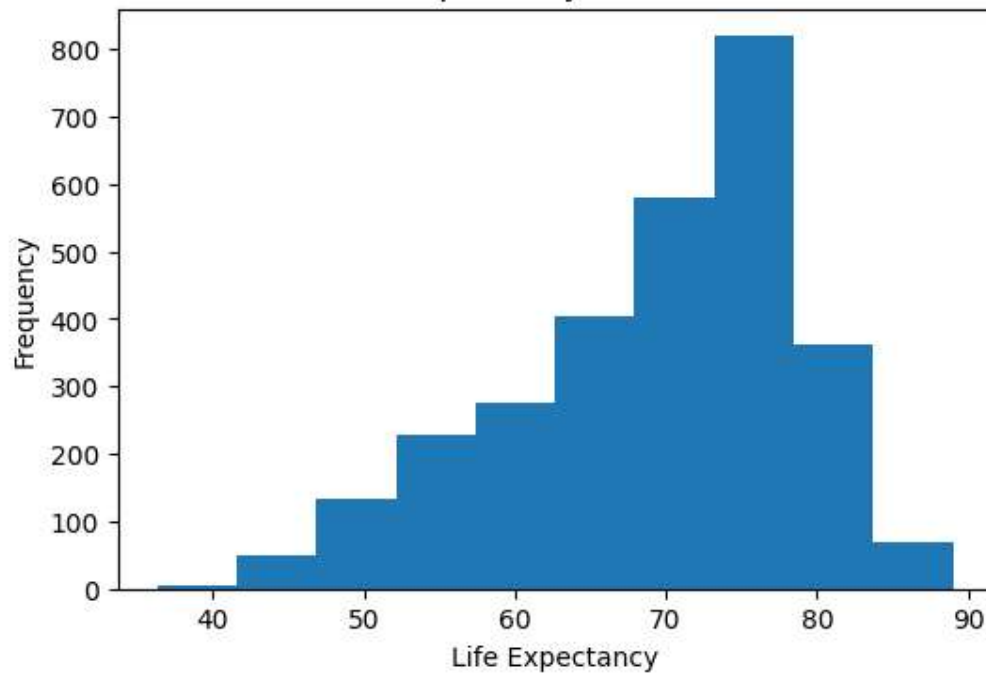
Output:

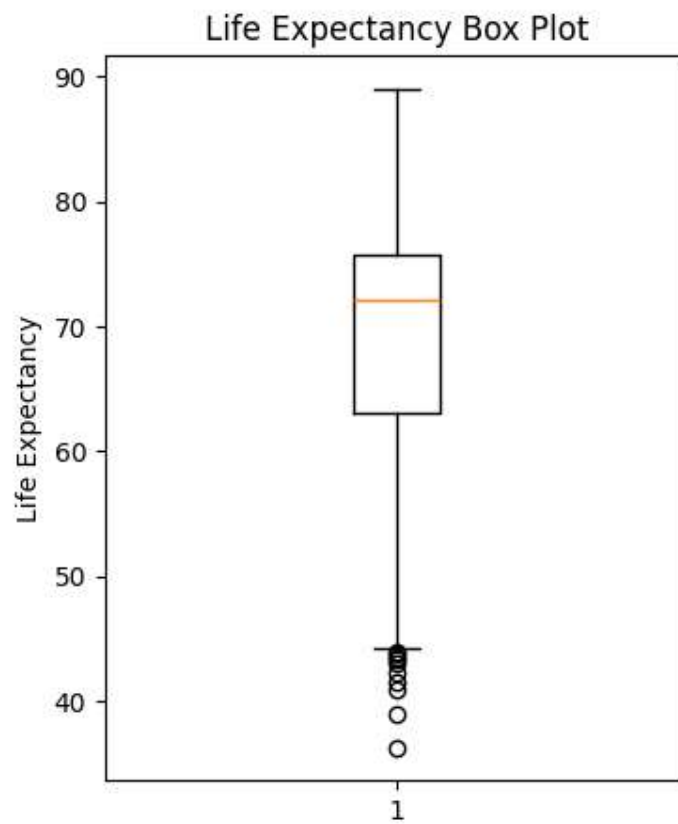
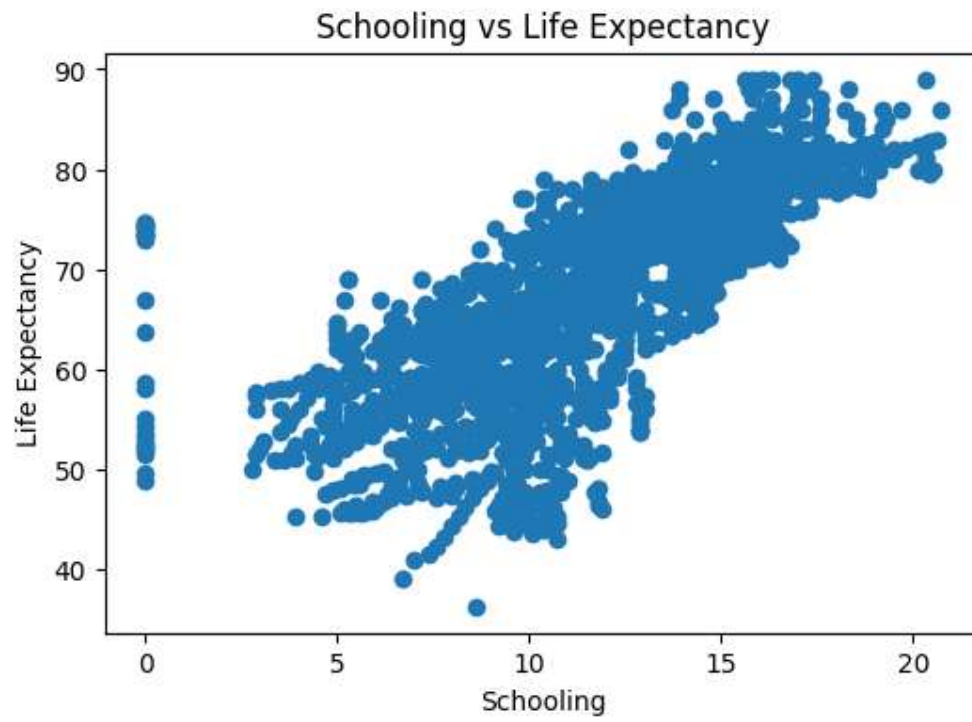


Development Status Distribution



Life Expectancy Distribution





Result: